

1. Administrative & Study Identification

Posting ID: 824312301445123609 **Brief Study Title:** Study of AZD9574 as Monotherapy and in Combination With Anti-cancer Agents in Participants With Advanced Solid Malignancies **Full Study Title:** A Modular Phase I/IIa, Open-label, Multi-centre Study to Assess the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Preliminary Efficacy of Ascending Doses of AZD9574 as Monotherapy and in Combination with Anti-cancer Agents in Patients with Advanced Solid Malignancies (CERTIS1) **Short Title/Acronym:** CERTIS1 **Protocol Number:** D8410C00001 **NCT Number:** NCT05417594 **Trial Status:** RECRUITING **Sponsor/Company Name:** AstraZeneca **Investigational Product:** AZD9574 (PARP1-selective BBB inhibitor), alone and in combination with Temozolomide (TMZ), T-DXd, Dato-DXd, or [11C]AZ14193391 **Phase of Development:** Phase I/IIa (PHASE1, PHASE2) **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of AZD9574 as monotherapy and in combination with anti-cancer agents at increasing dose levels in participants with advanced solid malignancies, and to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To assess the pharmacokinetics (PK), pharmacodynamics (PD), and preliminary efficacy (e.g., Objective Response Rate, Progression-Free Survival) of AZD9574 individually and in combination regimens.

2.2 Background & Rationale AZD9574 is a next-generation, PARP1-selective targeted therapy. The hypothesis is that AZD9574 will offer a more efficacious and less toxic cancer treatment compared with currently approved PARP inhibitors. A unique property of AZD9574 is its ability to readily cross the blood-brain barrier (BBB), which may provide a much-needed new treatment option for patients with central nervous system (CNS) malignancies (such as IDH1/2-mutant gliomas) and solid tumors with brain metastases.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation and expansion cohorts per module) **Blinding:** Open-label **Randomization:** Non-randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration Targeted Enrollment:

Approximately 695 participants **Number of Sites:** Multi-centre (Global) **Estimated Study Start:** 24-Jun-2022 **Estimated Primary Completion:** 11-Aug-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Histologically or cytologically confirmed advanced/metastatic solid malignancies (including breast, ovarian, castration-resistant prostate cancer [CRPC], pancreatic cancer, or IDH1/2-mutant glioma) that have recurred or progressed. 3. Must have evaluable disease with at least one measurable lesion per RECIST 1.1 (or RANO for glioma) measuring ≥ 10 mm in the longest diameter. 4. Evidence of predicted loss of function germline or tumor mutations (e.g., BRCA1, BRCA2, PALB2, RAD51C, RAD51D) depending on the specific module/cohort.

4.2 Exclusion Criteria 1. History of uncontrolled seizures, need for >2 concurrent antiepileptic drugs, severe brain injury, or stroke. 2. Any evidence of severe or uncontrolled systemic diseases including active bleeding diatheses, or active infections (including HBV, HCV, and HIV). 3. Prior treatment with a PARP inhibitor (for specific PARPi-naïve cohorts), or more than one prior line of alkylating chemotherapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ascending doses of AZD9574 administered orally in modular cohorts. Combination modules include AZD9574 plus temozolomide (drug name: temozolomide; trade name: Temodar; ATC code: L01AX03), T-DXd (intravenous), Dato-DXd (intravenous), or [^{11}C]AZ14193391. **Route of Administration:** Oral (AZD9574, TMZ) and Intravenous (T-DXd, Dato-DXd, Tracer). **Duration of Treatment:** Continuous treatment until objective disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for advanced cancer. **Prohibited:** Concurrent investigational agents, prior PARP inhibitors (for select modules), and recent administration of bevacizumab (within the last 6 months).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor Molecular Profiling (e.g., BRCA status), Baseline Imaging
Baseline	Day 0	Module Assignment, First	

Dose of Study Drug, PK/PD Baseline Sampling | | **Interim** | Varies
| PK/PD Blood Draws, Safety Labs, Efficacy Assessment (RECIST
1.1/RANO Tumor Imaging), AE Monitoring | | **End of Study** | N/A |
Treatment Discontinuation, Final Vital Signs, Exit Interview, 3-6
Month Safety/Contraception Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Incidence of Adverse Events (AEs), and Serious Adverse Events (SAEs):** Assessment of safety and tolerability by evaluating AEs, SAEs, and Dose-Limiting Toxicities (DLTs) to determine the MTD/RP2D. 2. **Changes from baseline in laboratory findings, electrocardiograms (ECGs), and vital signs:** Continuous clinical and laboratory assessments graded per NCI CTCAE criteria. 3. **Change from baseline Eastern Cooperative Oncology Group performance status (ECOG PS):** Measured on a scale of 0 to 4, where '0' indicates fully active and '4' indicates completely disabled.

7.2 Secondary & Exploratory Outcomes 1. **Pharmacokinetics (PK):** Area Under the Curve (AUC), Maximum plasma concentration (C_{max}), and Time to reach maximum plasma concentration (t_{max}) of AZD9574. 2. **Preliminary Efficacy:** Objective Response Rate (ORR) and Progression-Free Survival (PFS) assessed per RECIST 1.1 and/or RANO guidelines.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of hematological toxicities (e.g., neutropenia, thrombocytopenia) and non-hematological AEs via physical exams and labs. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Internal Dose Escalation Committee (DEC) reviews safety and DLT data prior to advancing to the next dose cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients who receive at least one dose), Efficacy Analysis Set (patients with measurable baseline disease and ≥ 1 post-baseline assessment). **Sample Size Justification:** Targeted enrollment of 695 participants. The sample size is driven by the traditional phase I dose-escalation rules and adequate powering for early efficacy signals in the expansion cohorts. **Handling of Missing Data:** Standard oncology guidelines apply; patients without post-baseline tumor assessments are considered non-responders. Safety data is typically not imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Risk-based monitoring utilizing both routine onsite visits and centralized remote monitoring of critical data points. **Audit Trail:** Secure eCRF entry, continuous data clarification query generation, and central laboratory verification for tumor mutation profiles.

11. Study Identifiers & Governance

Primary ID: D8410C000001 **Registry Number:** NCT05417594 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** CERTIS1 **Official Regulatory Title:** A Modular Phase I/IIa, Open-label, Multi-centre Study to Assess the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Preliminary Efficacy of Ascending Doses of AZD9574 as Monotherapy and in Combination with Anti-cancer Agents in Patients with Advanced Solid Malignancies

12. Clinical Framework (Oncology Specific)

Disease Areas / Primary Condition: Advanced Solid Malignancies **Preferred UMLS Name:** Advanced Solid Malignant Neoplasm **Preferred Name:** Advanced Malignant Solid Neoplasm **Semantic Type:** Neoplastic Process **Definition:** A malignant solid neoplasm that has spread extensively to other anatomic sites or is no longer responding to treatment. **MeSH Code:** N/A **SNOMED ID:** 39958004 **SNOMED Hierarchy:** 785317006 | Advanced solid malignant neoplasm | Advanced solid malignant neoplasm (disorder) **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic or locally advanced disease **Medical Methodology:** Targeted PARP1-selective inhibition with BBB penetration **Optimization Target:** Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Sequential dose-escalation (Part A) followed by dose-expansion (Part B) across multiple combinations and monotherapy modules **Education Protocol (HCP):** Site initiation training, specific focus on managing dose-limiting hematological toxicities **Education Protocol (Patient):** Contraception requirement counseling (highly effective methods required up to 6 months post-treatment) and study diary training **Quality Audit Mechanism:** Centralized imaging reviews (RECIST 1.1/RANO) and central biomarker/genetic testing validation

14. Observation & Follow-Up Schedule

Total Duration: Dependent on disease progression; includes \$ \approx \$ 3-6 months post-treatment safety follow-up

Onsite Visit Cadence: Intensive monitoring during Cycle 1, shifting to routine day 1 of each subsequent cycle

Remote Monitoring Frequency:

Routine protocol-driven intervals

Checklist Review Interval:

Tumor imaging assessments typically performed every 8-12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]						